6. The solution of

$$y' + \frac{y}{x} = \frac{2}{x^2 y}, \qquad x \neq 0$$

is given by

A. $x^2y^2 + 4xy = C$

B. $x^2y^2 + 4x = C$

C. $xy^2 - 2x = C$

D. $x^2y^2 - 4x = C$

E. $xy^2 - 4x = C$

$$y' + \frac{1}{x} y = \frac{2}{x^2} y^{-1}$$

$$y' + P y = Q \cdot y^n$$

$$P = \frac{1}{x}, \quad Q = \frac{2}{x^2}, \quad n = -1$$

(Bernoulli)

Use $v = y^{1-n} = y^2$

$$y = v^{1/2}$$

$$y' = \frac{1}{2} v^{-1/2} \cdot \frac{dv}{dx} \implies \frac{1}{\cancel{2}} \cancel{v^{-1/2}} \frac{dv}{dx} + \frac{\cancel{1}\,2}{x} v^{1/2 + 1/2} = \frac{\cancel{2}\,4}{x^2} \cancel{v^{-1/2}}$$

$$\frac{dv}{dx} + \frac{2}{x} v^1 = \frac{4}{x^2} \qquad v' + Pv = Q$$

$P = 2/x$, $Q = 4/x^2$ $\quad \rho = e^{\int P dx} = e^{\int 2/x \, dx} = e^{2\ln x + c}$

pick $c = 0$ $\implies$ pick $\boxed{\rho = e^{\ln(x^2)} = x^2}$

$$v' + Pv = Q \longrightarrow \rho v' + \rho P v = \rho Q \quad \cdots \quad \rho v = \int \rho \cdot Q + C$$

$$x^2 v = \int x^2 \cdot (4/x^2)\, dx + C = \int 4 dx + C$$

$$= 4x + C$$

$$x^2 v = 4x + C \xrightarrow{(v = y^2)} x^2 y^2 = 4x + C$$

$$\implies \boxed{x^2y^2 - 4x = C}$$

9. According to the method of undetermined coefficients, what is the proper form of a particular solution Y to the following differential equation?

$$y^{(4)} - 4y'' = 24t^2 - 4 - 3te^t.$$

A. $Y(t) = At^2 + Bte^t$.

B. $Y(t) = At^2 + Bt + C + Dte^t + Ee^t$.

C. $Y(t) = At^3 + Bt^2 + Ct + D + Ete^t + Fe^t$.

D. $Y(t) = At^4 + Bt^3 + Ct^2 + Dte^t + Ee^t$.

E. None of the above.

General solution is $\boxed{y_c} + \boxed{y_p}$

y_c is solution for

$y^{(4)} - 4y'' = 0$

Char. eq. $r^4 - 4r^2 = 0$

$r^2(r^2-4) = r^2(r-2)(r+2)$

$r = 0, 0, 2, -2$ $\quad y_c = (c_1 + c_2 t) + c_3 e^{2t} + c_4 e^{-2t}$

For $f = 24t^2 - 4 - 3te^t$, trial solution

$y_p = A + Bt + Ct^2 + (D + Et)e^t$

remove duplication

new trial $y_p = At + Bt^2 + Ct^3 + (D+Et)e^t$

still duplication, so

new trial function $y_p = At^2 + Bt^3 + Ct^4 + (D+Et)e^t$

24. The general solution of $\mathbf{x}' = \begin{bmatrix} 3 & 5 \\ -1 & -1 \end{bmatrix} \mathbf{x}$ has the form

A. $c_1 e^t \begin{bmatrix} 2\cos t - \sin t \\ -\cos t \end{bmatrix} + c_2 e^t \begin{bmatrix} \cos t + 2\sin t \\ -\sin t \end{bmatrix}$

B. $c_1 e^t \begin{bmatrix} \cos t \\ 2\cos t - \sin t \end{bmatrix} + c_2 e^t \begin{bmatrix} -\sin t \\ \cos t - 2\sin t \end{bmatrix}$

C. $c_1 e^t \begin{bmatrix} 2\cos t + \sin t \\ -\cos t \end{bmatrix} + c_2 e^t \begin{bmatrix} \cos t + 2\sin t \\ \sin t \end{bmatrix}$

D. $c_1 e^t \begin{bmatrix} \cos t - 2\sin t \\ \sin t \end{bmatrix} + c_2 e^t \begin{bmatrix} 2\cos t + \sin t \\ -\cos t \end{bmatrix}$

E. None of the above.

$$= \boxed{e^{\cancel{t}} \begin{bmatrix} -2\cos\cancel{t} + \sin\cancel{t} \\ \cos\cancel{t} \end{bmatrix}} + \boxed{i e^{\cancel{t}} \begin{bmatrix} -\cos\cancel{t} - 2\sin\cancel{t} \\ \sin\cancel{t} \end{bmatrix}}$$

$$\underline{X}_1 = Re(e^{\lambda t}\underline{v}) \qquad \underline{X}_2 = Im(e^{\lambda t}\underline{v})$$

$$c_1 e^t \begin{bmatrix} +2\cos t - \sin t \\ -\cos t \end{bmatrix} + c_2 e^t \begin{bmatrix} +\cos t + 2\sin t \\ -\sin t \end{bmatrix}$$

24. The general solution of $\mathbf{x}' = \begin{bmatrix} 3 & 5 \\ -1 & -1 \end{bmatrix} \mathbf{x}$ has the form

A. $c_1 e^t \begin{bmatrix} 2\cos t - \sin t \\ -\cos t \end{bmatrix} + c_2 e^t \begin{bmatrix} \cos t + 2\sin t \\ -\sin t \end{bmatrix}$

B. $c_1 e^t \begin{bmatrix} \cos t \\ 2\cos t - \sin t \end{bmatrix} + c_2 e^t \begin{bmatrix} -\sin t \\ \cos t - 2\sin t \end{bmatrix}$

C. $c_1 e^t \begin{bmatrix} 2\cos t + \sin t \\ -\cos t \end{bmatrix} + c_2 e^t \begin{bmatrix} \cos t + 2\sin t \\ \sin t \end{bmatrix}$

D. $c_1 e^t \begin{bmatrix} \cos t - 2\sin t \\ \sin t \end{bmatrix} + c_2 e^t \begin{bmatrix} 2\cos t + \sin t \\ -\cos t \end{bmatrix}$

E. None of the above.

$A = \begin{bmatrix} 3 & 5 \\ -1 & -1 \end{bmatrix}$

e.vals:

$\det(A - \lambda I)$

$= \begin{vmatrix} 3-\lambda & 5 \\ -1 & -1-\lambda \end{vmatrix}$

$= (3-\lambda)(-1-\lambda) + 5$

$= -3 + \lambda - 3\lambda + \lambda^2 + 5$

$= \lambda^2 - 2\lambda + 2$

$\lambda = \dfrac{2 \pm \sqrt{4-8}}{2} = \dfrac{2 \pm \sqrt{-4}}{2} = \boxed{1 \pm i}$ conjugate root pair (mult. 1)

Find eigenvector for $\lambda = 1+i$, Solve

$\left(A - (1+i)I\right)\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} 2-i & 5 \\ -1 & -2-i \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} 2-i & 5 & 0 \\ -1 & -2-i & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 2+i & 0 \\ 2-i & 5 & 0 \end{array}\right]$ $a + (2+i)b = 0$, $a = -(2+i)b$

$\longrightarrow (R_2 \to R_2 - (2-i)R_1) \left[\begin{array}{cc|c} \boxed{1} & 2+i & 0 \\ 0 & 0 & 0 \end{array}\right]$

so eigenvector for $\lambda = 1+i$ are $\begin{bmatrix} (-2-i)b \\ b \end{bmatrix} = b\begin{bmatrix} -2-i \\ 1 \end{bmatrix}$

pick $\underline{v} = \begin{bmatrix} -2-i \\ 1 \end{bmatrix}$

Since we have conjugate pair, we can can find X_1, X_2 by making

$$X_1 = \mathrm{Re}\left(e^{\lambda t}\underline{v}\right), \quad X_2 = \mathrm{Im}\left(e^{\lambda t}\underline{v}\right)$$

$\lambda = 1+i$, $\underline{v} = \begin{bmatrix} -2-i \\ 1 \end{bmatrix}$

$$e^{\lambda t}\underline{v} = e^{(1+i)t}\begin{bmatrix}-2-i\\ 1\end{bmatrix} \qquad e^{ix} = \cos x + i\sin x$$

$$= e^{t}(\cos t + i\sin t)\left(\begin{bmatrix}-2\\ 1\end{bmatrix} + i\begin{bmatrix}-1\\ 0\end{bmatrix}\right)$$

$$= e^{t}\left(\begin{bmatrix}-2\cos t\\ \cos t\end{bmatrix} + \overset{-1}{\cancel{i^2}}\begin{bmatrix}-\sin t\\ 0\end{bmatrix}\right)$$

$$+ e^{t}\left(i\begin{bmatrix}-\cos t\\ 0\end{bmatrix} + i\begin{bmatrix}-2\sin t\\ \sin t\end{bmatrix}\right)$$

$$= \boxed{e^{t}\begin{bmatrix}-2\cos t + \sin t\\ \cos t\end{bmatrix}} + i\boxed{e^{t}\begin{bmatrix}-\cos t - 2\sin t\\ \sin t\end{bmatrix}}$$

$$\mathbf{X}_1 = \mathrm{Re}(e^{\lambda t}\underline{v}) \qquad \mathbf{X}_2 = \mathrm{Im}(e^{\lambda t}\underline{v})$$

gen solution $c_1\mathbf{X}_1 + c_2\mathbf{X}_2$

11. Let $y(x)$ be the solution of the initial value problem

$$\underline{y'' + y = x,}$$
$$y(0) = 1, \quad y'(0) = 2.$$

Then $y(\pi)$ is equal to

A. $y(\pi) = \pi$

B. $y(\pi) = 2\pi$

C. $y(\pi) = \pi - 1$

D. $y(\pi) = 2\pi + 1$

E. $y(\pi) = \pi + 1$

need $y = y_c + y_p$ account for $f(x) = x \neq 0$

solution for homog. eq. $y'' + y = 0$

$y'' + y = 0 \longrightarrow r^2 + 1 \quad r = \pm i$

$\underline{y_c = c_1 \cos(x) + c_2 \sin(x)}$

step 2 : trial function

$y_p = A + Bx$

$$\left.\begin{array}{l} y_p = A + Bx \\ y_p' = B \\ y_p'' = 0 \end{array}\right\} \quad y_p'' + y_p = \underline{A + Bx} = \underline{x}$$

$A = 0 \quad B = 1.$

so $\underline{y_p = x}$

so general solution is

$y = y_c + y_p = c_1 \cos(x) + c_2 \sin(x) + x$

$y(0) = 1$

$y'(0) = 2$

$1 = y(0) = c_1 \longrightarrow \underline{c_1 = 1}$

$y' = -\overset{1}{\cancel{c_1}} \sin(x) + c_2 \cos(x) + 1$

$2 = y'(0) = c_2 + 1 \Longrightarrow \underline{c_2 = 1.}$

particular solution is

$y = \cos(x) + \sin(x) + x$

$y(\pi) = -1 + 0 + \pi = \pi - 1$

The differential equation $y'' + (\tan x)y' = 0$ has two linearly independent solutions $y_1(x) = \sin x$ and $y_2(x) = 1$. If we use the method of Variation of Parameters to find a particular solution $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$ for the corresponding nonhomogeneous equation

$y_1 = \sin x$

$$y'' + (\tan x)y' = \cos^2 x \qquad y_2 = 1$$

then which of the following is a possible function $u_1(x)$?

looking for u_1, u_2, such that $y = u_1 y_1 + u_2 y_2$ is a solution for nonhomog. ODE.

This leads us to two equations for u_1', u_2':

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f \ (= \cos^2 x) \end{cases}$$

$$\begin{cases} u_1' \cdot \sin x + u_2' = 0 \\ u_1' \cdot \cos x + 0 = \cos^2 x \end{cases}$$

$u_1' \cdot \cos x = \cos^2 x$

$u_1' = \cos x$

Always we find that
$$\begin{cases} u_1' = -\dfrac{y_2 \cdot f}{W(y_1, y_2)} \\ u_2' = +\dfrac{y_1 \cdot f}{W(y_1, y_2)} \end{cases}$$

$$W(y_1, y_2) = W(\sin x, 1) = \begin{vmatrix} \sin x & 1 \\ \cos x & 0 \end{vmatrix} = -\cos x$$

So $u_1' = \dfrac{-y_2 \cdot f}{W} = \dfrac{-1 \cdot \cos^2 x}{-\cos x} = +\cos x$

$$u_1 = \int -\frac{y_2 f}{W}\, dx = \int \cos x \, dx = \sin x + c$$

General solution is $\quad u_1 y_1 + u_2 y_2$

$y_c = c_1 \sin(x) + c_2 \quad (\sin x + c)\sin x + u_2 \cdot 1$

$\sin^2 x + c \sin x + u_2$

full solution $\quad y_c + y_p = c_1 \sin x + c_2 + \sin^2 x + c \sin x + u_2$

$= c_1 \sin x + c_2 + \sin^2 x + u_2$

8. Which of the following is the general solution to $y'' + 4y = e^{2t} + 12\sin(2t)$?

$\boxed{\text{A. } y(t) = c_1\cos(2t) + c_2\sin(2t) + \frac{1}{8}e^{2t} - 3t\cos(2t)}$

B. $y(t) = c_1e^{2t} + c_2e^{-2t} + \frac{1}{4}t\,e^{2t} - 3t\cos(2t)$

C. $y(t) = c_1 + c_2e^{-4t} + \frac{1}{12}te^{2t} - 3t\cos(2t)$

D. $y(t) = c_1\cos(2t) + c_2\sin(2t) + \frac{1}{8}e^{2t} + 3\sin(2t)$

E. None of the above.

Need $y = y_c + y_p$, y_c is solution to $y'' + 4y = 0$.

$r^2 + 4 = 0 \Rightarrow r = \pm 2i$

so $y_c = c_1\cos(2t) + c_2\sin(2t)$. Trial y_p is

$y_p = Ae^{2t} + B\sin(2t) + C\cos(2t)$,

but this has duplication with y_c; so change

$$\boxed{y_p = Ae^{2t} + Bt\sin(2t) + Ct\cos(2t)}$$

$$y_p' = 2Ae^{2t} + B\sin(2t) + 2Bt\cos(2t) + C\cos(2t) - 2Ct\sin(2t)$$

$$y_p' = 2Ae^{2t} + (B - 2Ct)\sin(2t) + (2Bt + C)\cos(2t)$$

$$y_p'' = 4Ae^{2t} + -2C\sin(2t) + (2B - 4Ct)\cos(2t) + 2B\cos(2t) - (4Bt + 2C)\sin(2t)$$

$$\boxed{y_p'' = 4Ae^{2t} + (4B - \cancel{4Ct})\cos(2t) - (\cancel{4Bt} + 4C)\sin(2t)}$$

Then $y_p'' + 4y_p = \boxed{\because} + 4Ae^{2t} + \cancel{4Bt}\sin(2t) + \cancel{4Ct}\cos(2t)$

$= 8Ae^{2t} + 4B\cos(2t) - 4C\sin(2t)$

Need this to match $e^{2t} + 12\sin(2t)$,

$\Rightarrow A = 1/8$, $B = 0$, $C = -3$,

so $y_p = \frac{1}{8}e^{2t} - 3t\cos(2t)$.

gen. sol. $y = y_c + y_p = \boxed{c_1\cos(2t) + c_2\sin(2t) + \frac{1}{8}e^{2t} - 3t\cos(2t)}$